Customer Intelligence System for Strategic Business Decision Making

1. Problem Statement

Companies collect large volumes of customer data but often lack labeled information about customer behavior. Without segmentation, businesses cannot identify high-value customers, predict churn risk, or target marketing campaigns effectively.

This project aims to build an **AI-driven customer segmentation system using unsupervised machine learning algorithms**. The system identifies hidden patterns in customer purchasing behavior and groups customers into meaningful segments.

These segments can then be used to design **personalized marketing strategies, improve customer retention, and maximize revenue**.

2. Dataset Description

The dataset used in this project contains customer transaction and behavioral data. The dataset includes attributes related to purchasing behavior and customer engagement.

Key features used for segmentation include:

- **Recency (R):** Number of days since the last purchase
- **Frequency (F):** Number of purchases made by the customer
- **Monetary Value (M):** Total amount spent by the customer

These RFM features provide a strong representation of customer behavior and are widely used in marketing analytics.

3. Data Preprocessing

Before applying clustering algorithms, the dataset was cleaned and preprocessed.

Steps performed:

1. **Handling Missing Values**
   Missing values were checked and handled appropriately.

2. **Outlier Detection**
   Extreme values were analyzed to ensure clustering stability.

3. **Feature Engineering**
   RFM metrics were created to capture customer purchasing behavior.

4. **Feature Scaling**
   Since clustering algorithms rely on distance metrics, all features were standardized using **StandardScaler**.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the distribution and relationships within the dataset.

The following analyses were performed:

- Distribution of Recency, Frequency, and Monetary values
- Correlation analysis between customer features
- Identification of potential patterns in purchasing behavior

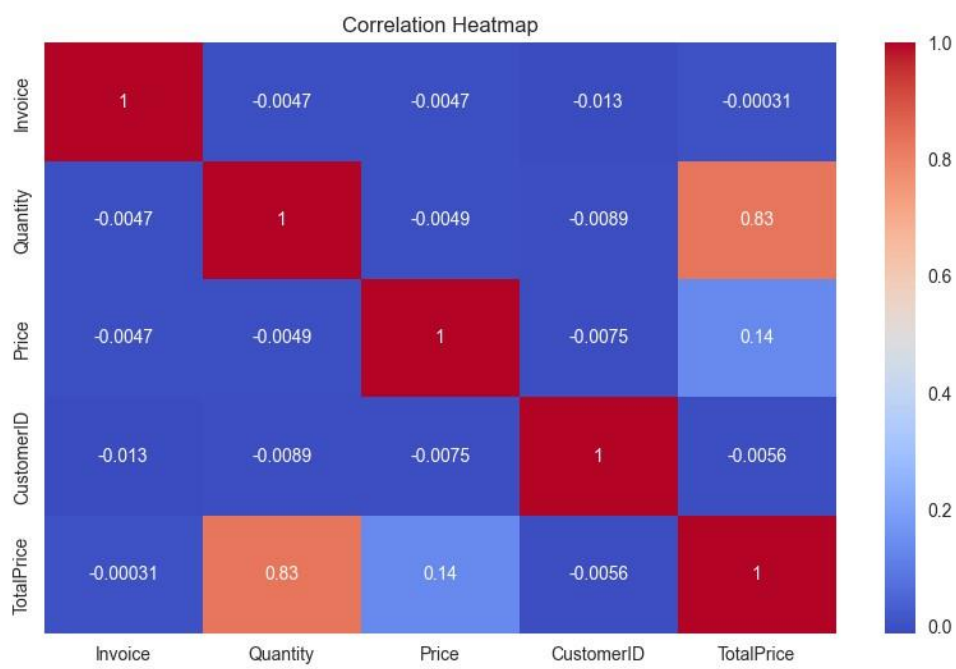


Figure 4.1 Correlation Heatm

5. Clustering Algorithms

To identify meaningful customer segments, multiple unsupervised clustering algorithms were implemented and compared.

Algorithms used:

- **K-Means Clustering**
- **Hierarchical Clustering**
- **DBSCAN**
- **Gaussian Mixture Model (GMM)**

Each algorithm groups customers based on similarity in RFM features.

Using multiple algorithms allows comparison of clustering performance and robustness.

6. Optimal Cluster Selection

Selecting the optimal number of clusters is a critical step in clustering analysis.

Two evaluation techniques were used:

- **Elbow Method**
- **Silhouette Score**
- **Davies-Bouldin Index**

The elbow method suggested that **4 clusters** provide the best balance between model complexity and cluster separation.

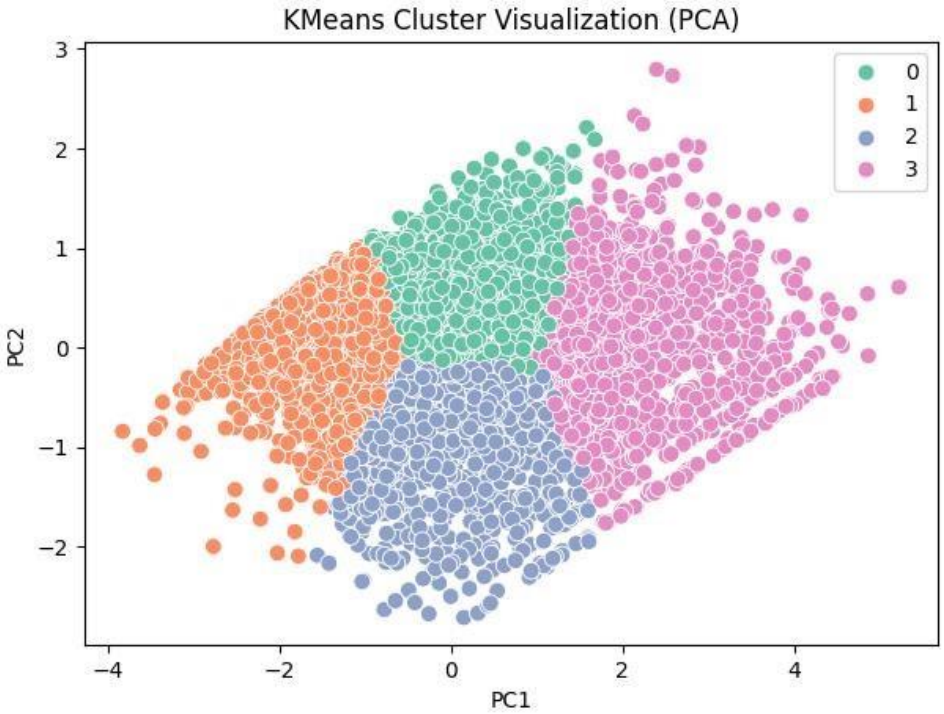
7. Dimensionality Reduction (PCA)

Since clustering occurs in high-dimensional space, **Principal Component Analysis (PCA)** was used to reduce the data to two dimensions for visualization.

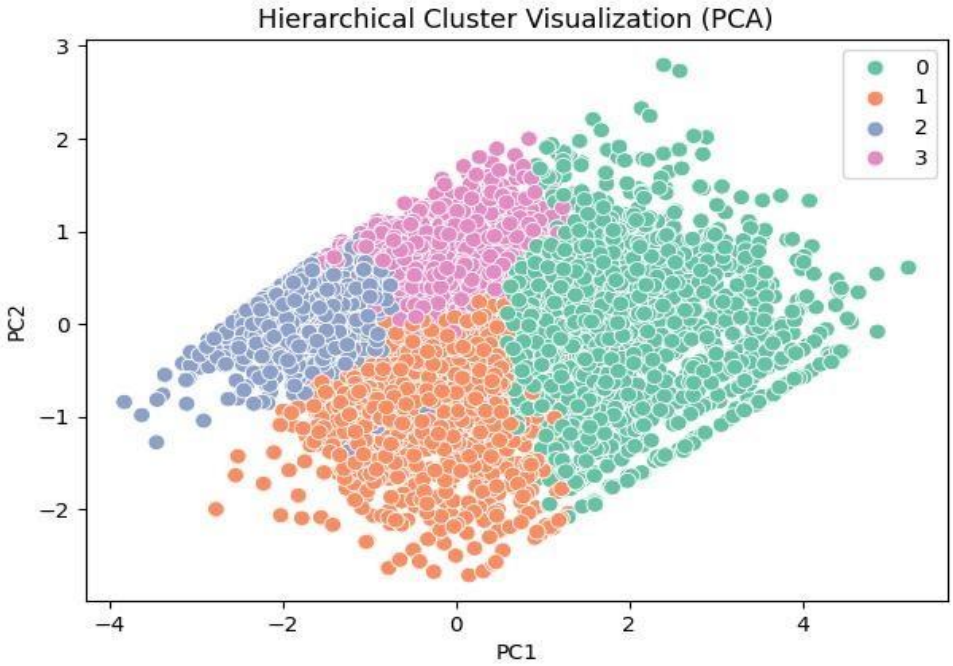
PCA transforms the original features into principal components that capture the maximum variance in the data.

This enables clear visualization of customer clusters.

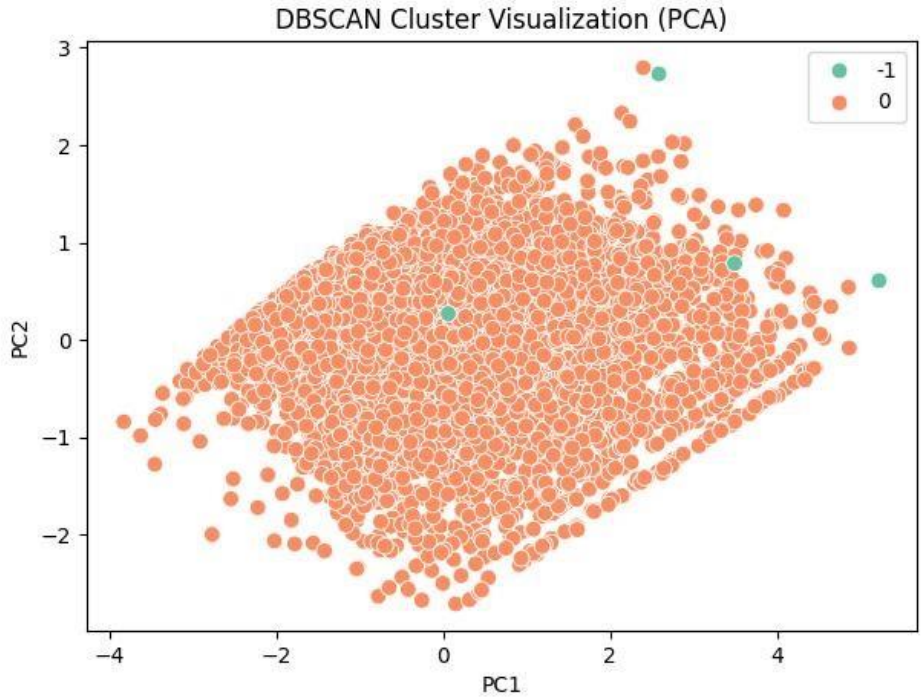
7.1 KMeans PCA clusters



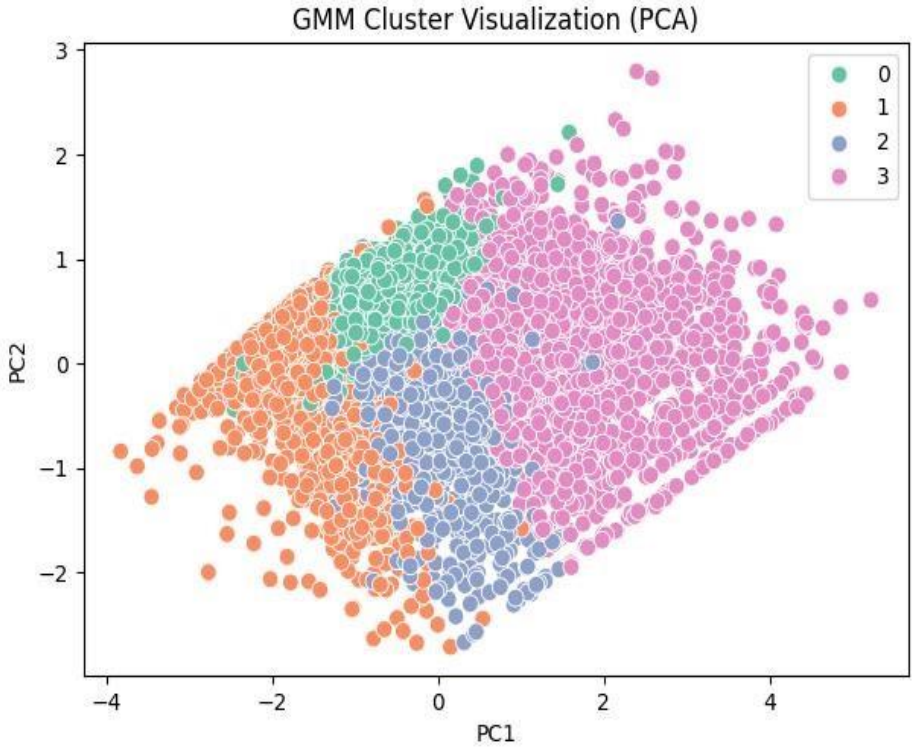
7.2 Hierarchical clusters



7.3 DBSCAN clusters



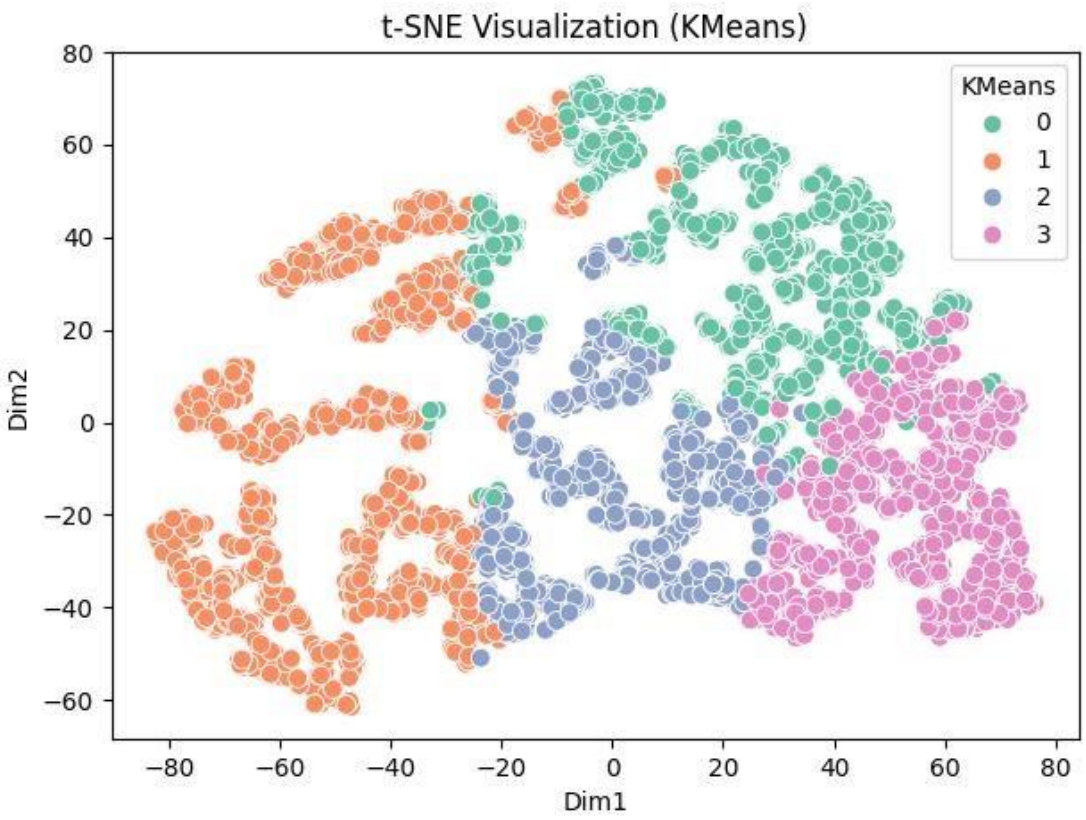
7.4 GMM clusters



8. t-SNE Visualization

To further analyze cluster separation, **t-SNE** was applied as an advanced dimensionality reduction technique. t-SNE is particularly effective for visualizing high-dimensional data in two dimensions.

8.1 t-SNE Graph



9. Cluster Interpretation

Cluster	Customer Type	Behavior
Cluster 0	High Value Customers	High spending and frequent purchases
Cluster 1	Budget Customers	Low spending but frequent visits
Cluster 2	Occasional Buyers	Low frequency purchases
Cluster 3	At-Risk Customers	Long time since last purchase

10. Business Insights

Based on the segmentation results, several actionable insights were discovered.

Key findings:

- **Cluster 0 represents high-value customers** who generate the largest revenue.
- **Cluster 1 includes price-sensitive buyers** who respond well to discounts.
- **Cluster 2 represents occasional shoppers** who require engagement strategies.
- **Cluster 3 represents at-risk customers** who have not purchased recently.

11. Marketing Strategy Recommendations

Customer Segment	Strategy
High Value Customers	Loyalty programs and premium offers
Budget Customers	Discount campaigns
Occasional Buyers	Personalized recommendations
At-Risk Customers	Retention campaigns

12. Conclusion

This project successfully implemented an end-to-end customer segmentation system using unsupervised learning techniques.

By applying clustering algorithms and dimensionality reduction techniques, meaningful customer segments were identified.

These insights can help businesses design targeted marketing strategies, improve customer retention, and maximize revenue.